

Homework 5

```
library(tidyverse)
library(mdsr)
library(dbplyr)
library(DBI)
```

```
# connect to the database which lives on a remote server maintained by
#   St. Olaf's IT department

library(RMariaDB)
con <- dbConnect(
  MariaDB(), host = "mdb.stolaf.edu",
  user = "ruser", password = "ruserpass",
  dbname = "flight_data"
)
```

On Your Own - Adapting 164 Code - HW

These problems are based on class exercises from SDS 164, so you've already solved them in R! Now we're going to try to duplicate those solutions in SQL (but with 2023 data instead of 2013).

```
# Read in 2013 NYC flights data
library(nycflights13)
flights_nyc13 <- nycflights13::flights
planes_nyc13 <- nycflights13::planes
```

1. Summarize carriers flying to MSP by number of flights and proportion that are cancelled (assuming that a missing arrival time indicates a cancelled flight). [This was #4 in 17_longer_pipelines.Rmd.]

```
# Original solution from SDS 164
flights_nyc13 |>
  mutate(carrier = fct_collapse(carrier, "Delta +" = c("DL", "9E"),
                                "American +" = c("AA", "MQ"),
                                "United +" = c("EV", "OO", "UA"))) |>

  filter(dest == "MSP") |>
  group_by(origin, carrier) |>
  summarize(n_flights = n(),
            num_cancelled = sum(is.na(arr_time)),
            prop_cancelled = mean(is.na(arr_time)))
```

```
# A tibble: 5 x 5
# Groups:   origin [3]
  origin carrier    n_flights num_cancelled prop_cancelled
  <chr>   <fct>         <int>         <int>         <dbl>
1 EWR    Delta +           598             10          0.0167
2 EWR    United +         1779            105          0.0590
3 JFK    Delta +          1095             41          0.0374
4 LGA    Delta +          2420             25          0.0103
5 LGA    American +        1293             62          0.0480
```

First duplicate the output above, then check trends in 2023 across all origins. Here are a few hints:

- use `flightdata` instead of `flights_nyc13`
- remember that `flights_nyc13` only contained 2013 and 3 NYC origin airports (EWR, JFK, LGA)
- `is.na` can be replaced with `CASE WHEN ArrTime IS NULL THEN 1 ELSE 0 END` or with `CASE WHEN cancelled = 1 THEN 1 ELSE 0 END`
- `CASE WHEN` can also be used replace `fct_collapse` # factor collapse

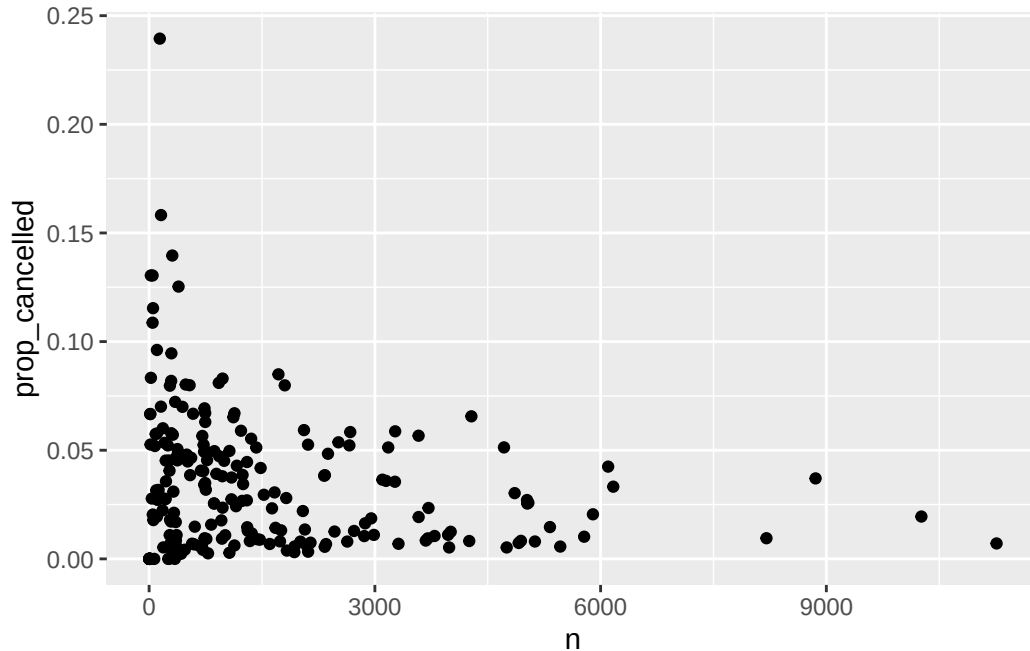
```
SELECT
  origin,
  Reporting_Airline AS carrier,
  COUNT(*) AS num_flights,
  SUM(CASE WHEN ArrTime IS NULL THEN 1 ELSE 0 END) AS num_cancelled,
  AVG(CASE WHEN cancelled = 1 THEN 1 ELSE 0 END) AS prop_cancelled
FROM flightdata
WHERE Year = 2023 AND dest = 'MSP'
GROUP BY origin, carrier
LIMIT 0, 20;
```

Table 1: Displaying records 1 - 10

origin	carrier	num_flights	num_cancelled	prop_cancelled
ABQ	DL	206	3	0.0146
ABR	OO	725	14	0.0193
ANC	AS	60	1	0.0167
ANC	DL	551	4	0.0073
ATL	9E	1	0	0.0000
ATL	DL	3145	18	0.0057
ATL	NK	643	10	0.0156
ATW	9E	159	2	0.0126
ATW	OO	668	5	0.0075
AUS	DL	1141	14	0.0123

2. Plot number of flights vs. proportion cancelled for every origin-destination pair (assuming that a missing arrival time indicates a cancelled flight). [This was #7 in 17_longer_pipelines.Rmd.]

```
# Original solution from SDS 164
flights_nyc13 |>
  group_by(origin, dest) |>
  summarize(n = n(),
            prop_cancelled = mean(is.na(arr_time))) |>
  filter(prop_cancelled < 1) |>
  ggplot(aes(n, prop_cancelled)) +
  geom_point()
```

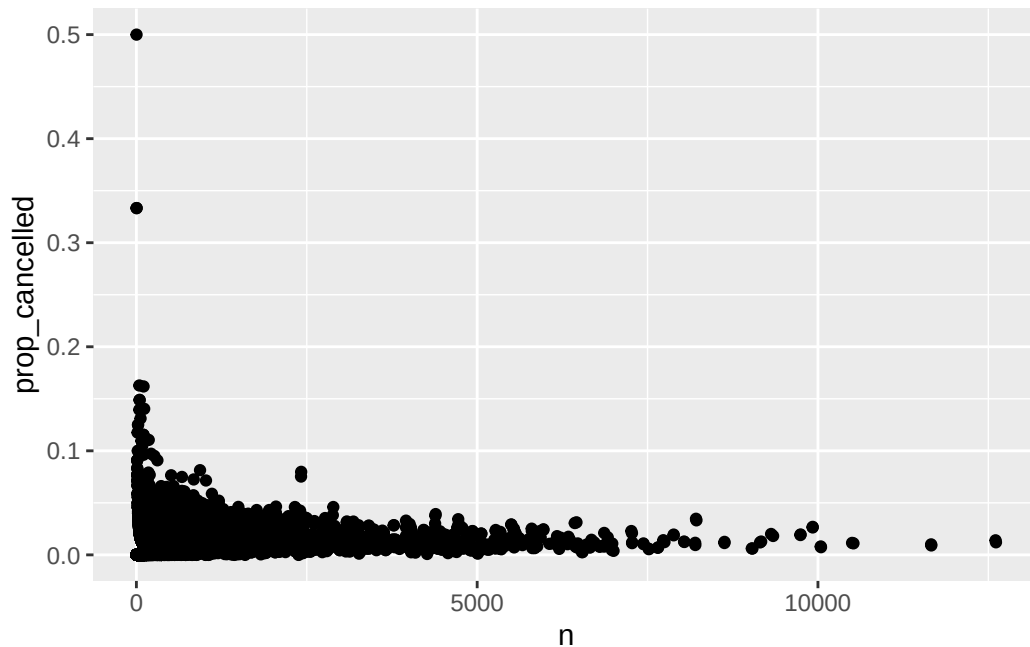


First duplicate the plot above for 2023 data, then check trends across all origins. Do all of the data wrangling in SQL. Here are a few hints:

- use `flightdata` instead of `flights_nyc13`
- remember that `flights_nyc13` only contained 2013 and 3 NYC origin airports (EWR, JFK, LGA)
- use an `sql` chunk and an `r` chunk
- include `connection =` and `output.var =` in your `sql` chunk header (this doesn't seem to work with `dbGetQuery()`...)

```
my_flights_data <- dbGetQuery(con, '
SELECT origin, dest,
       SUM(1) AS n,
       AVG(CASE WHEN cancelled = 1 THEN 1 ELSE 0 END) AS prop_cancelled
FROM flightdata
WHERE Year = 2023
GROUP BY origin, dest
HAVING prop_cancelled < 1;
')

ggplot(my_flights_data, aes(n, prop_cancelled)) +
  geom_point()
```



3. Produce a table of weighted plane age by carrier, where weights are based on number of flights per plane. [This was #6 in 26_more_joins.Rmd.]

```
# Original solution from SDS 164
flights_nyc13 |>
  left_join(planes_nyc13, join_by(tailnum)) |>
  mutate(plane_age = 2013 - year.y) |>
  group_by(carrier) |>
  summarize(unique_planes = n_distinct(tailnum),
            mean_weighted_age = mean(plane_age, na.rm =TRUE),
            sd_weighted_age = sd(plane_age, na.rm =TRUE)) |>
  arrange(mean_weighted_age)
```

A tibble: 16 x 4

	carrier	unique_planes	mean_weighted_age	sd_weighted_age
	<chr>	<int>	<dbl>	<dbl>
1	HA	14	1.55	1.14
2	AS	84	3.34	3.07
3	VX	53	4.47	2.14
4	F9	26	4.88	3.67
5	B6	193	6.69	3.29
6	00	28	6.84	2.41
7	9E	204	7.10	2.67

8	US	290	9.10	4.88
9	WN	583	9.15	4.63
10	YV	58	9.31	1.93
11	EV	316	11.3	2.29
12	FL	129	11.4	2.16
13	UA	621	13.2	5.83
14	DL	629	16.4	5.49
15	AA	601	25.9	5.42
16	MQ	238	35.3	3.13

First duplicate the output above for 2023, then check trends across all origins. Do all of the data wrangling in SQL. Here are a few hints:

- use flightdata instead of flights_nyc13
- remember that flights_nyc13 only contained 2013 and 3 NYC origin airports (EWR, JFK, LGA)
- you'll have to merge the flights dataset with the planes dataset
- you can use DISTINCT inside a COUNT()
- investigate SQL clauses for calculating a standard deviation
- you cannot use a derived variable inside a summary clause in SELECT

For bonus points, also merge the airlines dataset and include the name of each carrier and not just the abbreviation!

```
SELECT
  a.name,
  f.Reporting_Airline AS carrierID,
  COUNT(DISTINCT f.TAIL_NUMBER) AS unique_planes,
  AVG(2023 - p.year) AS mean_weighted_age,
  STDDEV_SAMP(2023 - p.year) AS sd_weighted_age
FROM flightdata AS f
JOIN planes AS p ON f.TAIL_NUMBER = p.tailnum
JOIN airlines AS a ON f.Reporting_Airline = a.carrier
GROUP BY carrier
ORDER BY mean_weighted_age ASC
LIMIT 0, 10;
```

Table 2: Displaying records 1 - 10

name	carrierID	unique_planes	mean_weighted_age	sd_weighted_age
Frontier Airlines Inc.	F9	156	5.1488	2.875346
Spirit Air Lines	NK	213	6.8567	3.858105

name	carrierID	unique_planes	mean_weighted_age	sd_weighted_age
Horizon Air	QX	47	7.8606	4.268428
PSA Airlines Inc.	OH	170	11.3136	5.682408
Alaska Airlines Inc.	AS	251	11.4766	6.752782
Envoy Air	MQ	272	11.8303	7.188317
Mesa Airlines Inc.	YV	137	12.2227	5.601301
SkyWest Airlines Inc.	OO	581	12.5643	6.851667
Southwest Airlines Co.	WN	820	12.7597	6.419219
Republic Airline	YX	204	12.7657	4.189681